

# 1 Detailed Metric Definition

## 1.1 Variables and Derived Maps

The dataset consists of  $C \times N = E \cdot S$  videos, where  $E$  experiments are each captured with  $C$  takes across  $S$  viewing angles. Each video is a tensor  $V \in \mathbb{R}^{H \times W \times T}$ , where  $H \times W$  denotes spatial resolution and  $T$  the number of frames.

From each video, a binary spatiotemporal activation map  $A_{\text{source}}^{\text{ST}} \in \{0, 1\}^{H \times W \times T}$  is derived from the greyscale signal, encoding where and when motion or activation occurs (see [1, Algo. 2] for details). Two further representations are derived from this map for use in the metrics:

- The **spatial activation map**  $A_{\text{source}}^{\text{SP}} \in \{0, 1\}^{H \times W}$  captures *where* any activation occurred across the full video.
- The **weighted spatial activation map**  $A_{\text{source}}^{\text{WS}} = \frac{1}{T} \sum_{t=1}^T A_{\text{source}; :, :, t}^{\text{ST}} \in [0, 1]^{H \times W}$  captures *where and how much* activation occurred, weighted by temporal frequency. Note that whether or not normalization is applied does not affect the resulting Weighted-Spatial-IoU score.

## 1.2 Basic Metric Definitions

Three IoU-based metrics are defined over the activation maps, and one pixel-level reconstruction metric over the raw video:

$$\text{Spatial-IoU} = \text{IoU}^{\text{SP}}(A_1^{\text{SP}}, A_2^{\text{SP}}) = \frac{|A_1^{\text{SP}} \cap A_2^{\text{SP}}|}{|A_1^{\text{SP}} \cup A_2^{\text{SP}}|} \in [0, 1] \quad (1)$$

$$\text{Spatiotemporal-IoU} = \text{IoU}^{\text{ST}}(A_1^{\text{ST}}, A_2^{\text{ST}}) = \sum_{t=1}^T \frac{1}{T} \frac{|A_{1, :, :, t}^{\text{ST}} \cap A_{2, :, :, t}^{\text{ST}}|}{|A_{1, :, :, t}^{\text{ST}} \cup A_{2, :, :, t}^{\text{ST}}|} \in [0, 1] \quad (2)$$

$$\text{Weighted-Spatial-IoU} = \text{IoU}^{\text{WS}}(A_1, A_2) = \frac{\sum_{i=1}^{HW} \min(A_{1;i}^w, A_{2;i}^w)}{\sum_{i=1}^{HW} \max(A_{1;i}^w, A_{2;i}^w)} \in [0, 1] \quad (3)$$

$$\text{MSE}(V_1, V_2) = \frac{1}{HWT} \|V_1 - V_2\|_F \in [0, 1] \quad (4)$$

where the MSE is computed as the mean over all frames of an experiment, with videos normalised to  $[0, 1]$ .

The metric values  $r_n^M$  for a single sample and the corresponding physical variation  $r_n^M$  are defined as:

$$v_n^{M_{\text{IoU}}} = \text{IoU}^M(A_{\text{GT};n}^M, A_{\text{Gen};n}^M) \quad (5)$$

$$v_n^{\text{MSE}} = \text{MSE}(V_{\text{GT};n}, V_{\text{Gen};n}) \quad (6)$$

$$r_n^{M_{\text{IoU}}} = \text{IoU}^M(A_{\text{GT};n}^M, A_{\text{GT}2;n}^M) \quad (7)$$

$$r_n^{M_{\text{MSE}}} = \text{MSE}(V_{\text{GT};n}, V_{\text{GT}2;n}) \quad (8)$$

## 1.3 Original Physics-IQ Score Aggregation

Each metric  $M \in \{\text{SP}, \text{ST}, \text{WS}, \text{MSE}\}$  is aggregated over the  $N$  evaluation videos into a mean score  $\mu^M$  and a physical variation ceiling  $\epsilon^M$ . The ceiling is computed by comparing the two ground-truth takes of each experiment, quantifying the irreducible trial-to-trial variability of the physical phenomena.

For the IoU metrics the subscores are defined as:

$$\mu^M = \frac{1}{N} \sum_{n=1}^N \text{IoU}^M(A_{\text{GT};n}^M, A_{\text{Gen};n}^M) \in [0, 1] \quad (9)$$

$$\epsilon^M = \frac{1}{N} \sum_{n=1}^N \text{IoU}^M(A_{\text{GT};n}^M, A_{\text{GT}2;n}^M) \in [0, 1] \quad (10)$$

$$s^M = \frac{\mu^M}{\epsilon^M} \in \mathbb{R}_+ \quad (11)$$

For MSE, lower is better, so the ceiling is subtracted rather than used as a divisor:

$$\mu^{\text{MSE}} = \frac{1}{N} \sum_{n=1}^N \text{MSE}(V_{\text{GT};n}, V_{\text{Gen};n}) \in [0, 1] \quad (12)$$

$$\epsilon^{\text{MSE}} = \frac{1}{N} \sum_{n=1}^N \text{MSE}(V_{\text{GT};n}, V_{\text{GT}2;n}) \in [0, 1] \quad (13)$$

$$s^{\text{MSE}} = \mu^{\text{MSE}} - \epsilon^{\text{MSE}} \in [-1, 1] \quad (14)$$

#### 1.4 Stable Physics-IQ Score

For the original composite Physics-IQ score the three IoU sub-scores are averaged and the MSE penalty is subtracted to produce a raw composite score, which is then clipped to  $[0, 1]$ :

$$s_{\text{Physics-IQ}} = c_{[0,1]} \left( \frac{1}{3} (s^{\text{SP}} + s^{\text{ST}} + s^{\text{WS}}) - s^{\text{MSE}} \right) \quad (15)$$

where each subscore is normalised by the physical variation ceiling, representing the typical deviation between independent second takes of the same experiment. The structural flaw in Eq. 15 is that the scores for each metric, Spatial, Spatiotemporal and weighted spatial IoU ( $s^{\text{SP}}, s^{\text{ST}}, s^{\text{WS}} \in [0, \infty)$ ), are unbounded: a single exceptional subscore can dominate the composite irrespective of performance on the remaining metrics, directly contradicting the design intent of Motamed et al. [1] that “no metric should be assessed in isolation.” By construction, a subscore of 1 for the positive metrics indicates that the generated videos match the ground truth as well as a second take would; scores above 1 indicate that estimated ceiling performance has been surpassed, which the outer  $c_{[0,1]}$  does not prevent from inflating the average before aggregation.

As Physics-IQ is designed to assess physical understanding relative to natural scene variability, not to reward performance beyond second-take realism. We therefore enforce a performance ceiling at the physical variation by clipping each subscore individually before aggregation resulting in the *Physics-IQ stable* composite score:

$$s_{\text{Physics-IQ stable}} = c_{[0,1]} \left( \frac{1}{3} (c_{[0,1]}(s^{\text{SP}}) + c_{[0,1]}(s^{\text{ST}}) + c_{[0,1]}(s^{\text{WS}})) - c_{[0,1]}(s^{\text{MSE}}) \right) \quad (16)$$

The symmetry here is one of design intent rather than mathematical range. For  $s^{\text{SP}}, s^{\text{ST}}, s^{\text{WS}} \geq 1$  indicate better-than-ceiling performance and are clipped to 1. For  $s^{\text{MSE}} \leq 0$  indicate better-than-ceiling pixel similarity and are clipped to 0. The symmetry here is one of design intent rather than mathematical range. Per-metric clipping ensures that no individual subscore can contribute beyond its intended  $\frac{1}{3}$  share of the composite, while preserving full sensitivity in the practically relevant regime of below-ceiling performance. This correction is principled regardless of empirical impact; where it additionally affects model rankings, this reflects the degree to which the original formula was distorted by subscore dominance.<sup>1</sup>

<sup>1</sup>At the time of writing the highest scores for the original Physics-IQ benchmark is at  $62.6 \approx \frac{100}{3} (\frac{43.9}{66.4} + \frac{33.9}{53.2} + \frac{33.9}{56.9}) - (0.5 - 0.2)$  [2]. Therefore video generative models are not close yet in any metric towards hitting the performance ceiling.

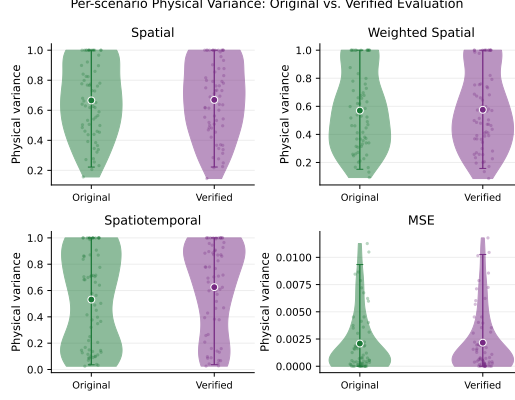


Figure 1: Visualization of the physical variance distribution  $r_n^M$  per scenario obtained using the verified and the original ground truth. The results clearly indicate that the result is not gaussian distributed supporting the notion that the mean physical variance potentially downweights the influence of samples with a low physical variance.

### 1.5 Drawbacks of the Original Score

- The score is not defined for a single sample but only over the entire dataset.  $\rightarrow$  This makes it unclear on which samples a model performs well and on which samples it does not perform well.
- The mean aggregation for the physical variation leads to smaller values contributing less to the overall score.  $\rightarrow$  Samples that have a smaller physical variation and in theory also smaller scores contribute less to the overall score.
- The original unclipped score could have an overflow for sub-score values greater 1 (or smaller than 0 for MSE).  $\rightarrow$  A single score very high score can dominate the Physics-IQ score.

### 1.6 Sample-Level Physics-IQ Verified Score

We propose a principled physics-iq score operating on the sample level  $i \in \{1, \dots, N\}$  over the entire dataset. The aggregation of each score is performed using the arithmetic mean so that improvements across every single metric are clearly attributed in the per sample score.

Additionally, we change the interpretation of the MSE which now captures how many times the generated MSE score is larger than that of the physical variation.

$$s_n^{\text{Physics-IQ verified}} = \frac{1}{4} \left( \underbrace{c_{[0,1]} \left( \frac{r_n^{\text{MSE}}}{v_n^{\text{MSE}}} \right)}_{s_n^{\text{MSE verified}}} + \sum_{M_{\text{IoU}} \in \{\text{SP, ST, WS}\}} \underbrace{c_{[0,1]} \left( \frac{v_n^{M_{\text{IoU}}}}{r_n^{M_{\text{IoU}}}} \right)}_{s_n^{M_{\text{IoU}} \text{ verified}}} \right) \quad (17)$$

The final Physics-IQ Verified score is the arithmetic mean across all samples:  $s^{\text{Physics-IQ verified}} = \frac{1}{N} \sum_{n=1}^N s_n^{\text{Physics-IQ verified}}$ .

The subscores for each metric  $s^{M \text{ verified}}$  over the entire dataset for our verified scores are obtained by summing over all samples in an identical fashion.

## References

- [1] Saman Motamed, Laura Culp, Kevin Swersky, Priyank Jaini, and Robert Geirhos. Do generative video models understand physical principles? In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 948–958, 2026.
- [2] Jianhao Yuan, Xiaofeng Zhang, Felix Friedrich, Nicolas Beltran-Velez, Melissa Hall, Reyhane Askari-Hemmat, Xiaochuang Han, Nicolas Ballas, Michal Drozdal, and Adriana Romero-Soriano. Inference-time physics alignment of video generative models with latent world models. *arXiv preprint arXiv:2601.10553*, 2026.